

Praktikum

Design of Wind Farms

Lecture 3:
Wake Behavior, Wind Farm Control

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Syllabus

No.	Week	Date (Wed)	9.30–10.15	10.30–11.15	11.30–12.15
1	14	15/04/26	Overview of Course Lecture 1: Background and Motivations	Lecture 2: Development of a Wind Farm Project	Lecture 3: Wake Behavior, Wind Farm Control Assignment 1: Basics and Recapitulation
2	15	22/04/26	Lecture 4: Wake Models, FLORIS	Assignment 1: Solution Assignment 2: Jensen's Wake Model and Wake Superposition in Wind Farms	Tutorial 1: FLORIS Installation and Setup
3	16	29/04/26	Lecture 5_1: LandBOSSE	Tutorial 2: LANDBOSSE Installation and Setup	Assignment 2: Solution Assignment 3: Wind Farm Analysis Using FLORIS
5	17	06/05/26	Lecture 5_2: Financial Analysis	Assignment 3: Solution	Assignment 4: Wind Farm Cost Analysis Using FLORIS + LandBOSSE)
6	18	13/05/26	Lecture 6: Layout Optimization	Assignment 4: Solution	Assignment 5: Layout Optimization
7	19	20/05/26	Assignment 5: Solution		Final group assignment: Introduction, Finalizing Groups, Q&A
8	20	27/05/26	Final group assignment: Design Process, Q&A, Writing Report		
9	21	03/06/26	No Class		
10	22	10/06/26	Final group assignment: Design Process, Q&A, Writing Report		
11	28	08/07/26	No Class Deadline: Electronic report submission until Tuesday, 07/07/2026, 23:59h		
12	29	15/07/26	Final presentation – 8:30h (Room 2701M)		

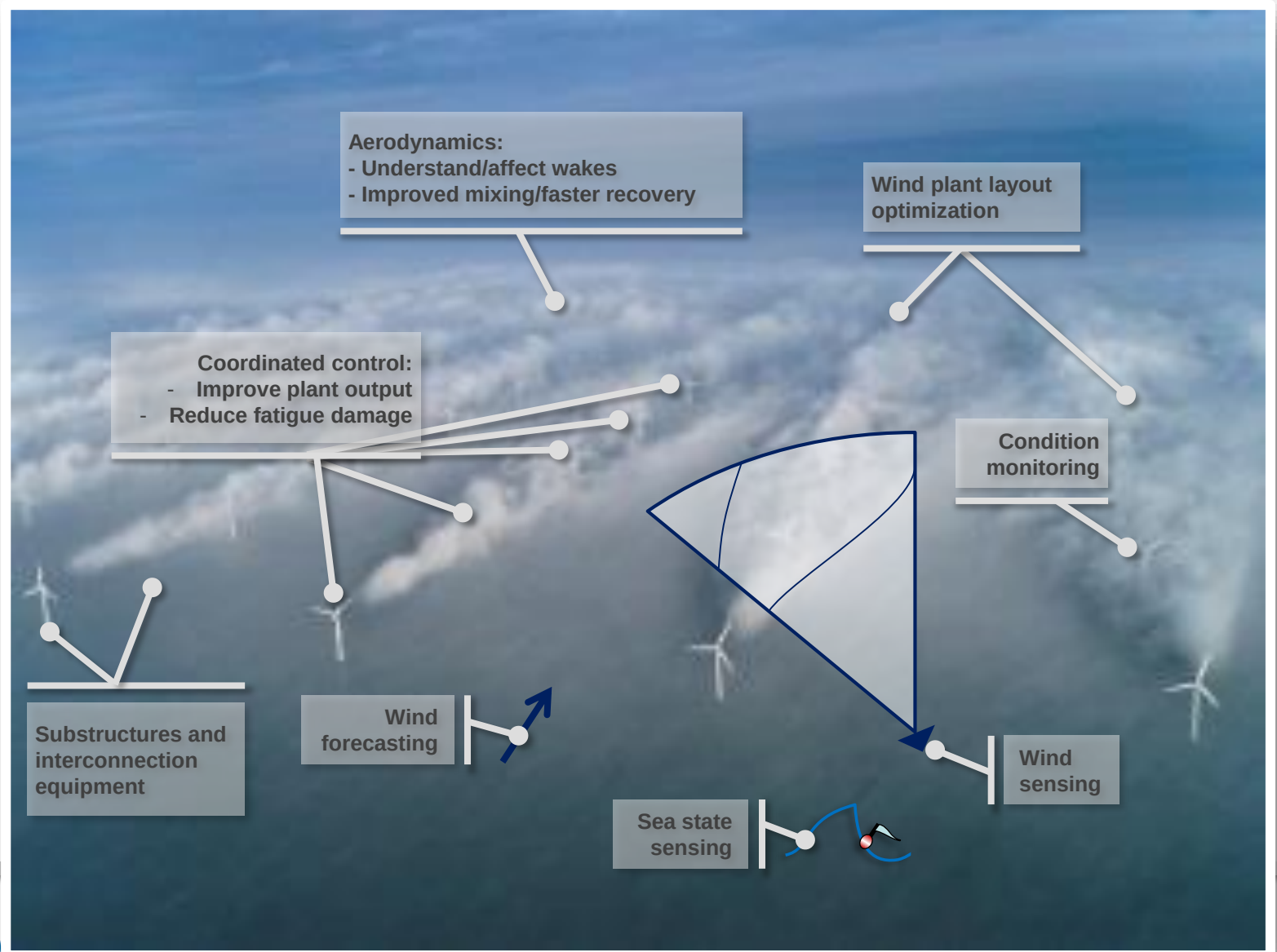


Agenda

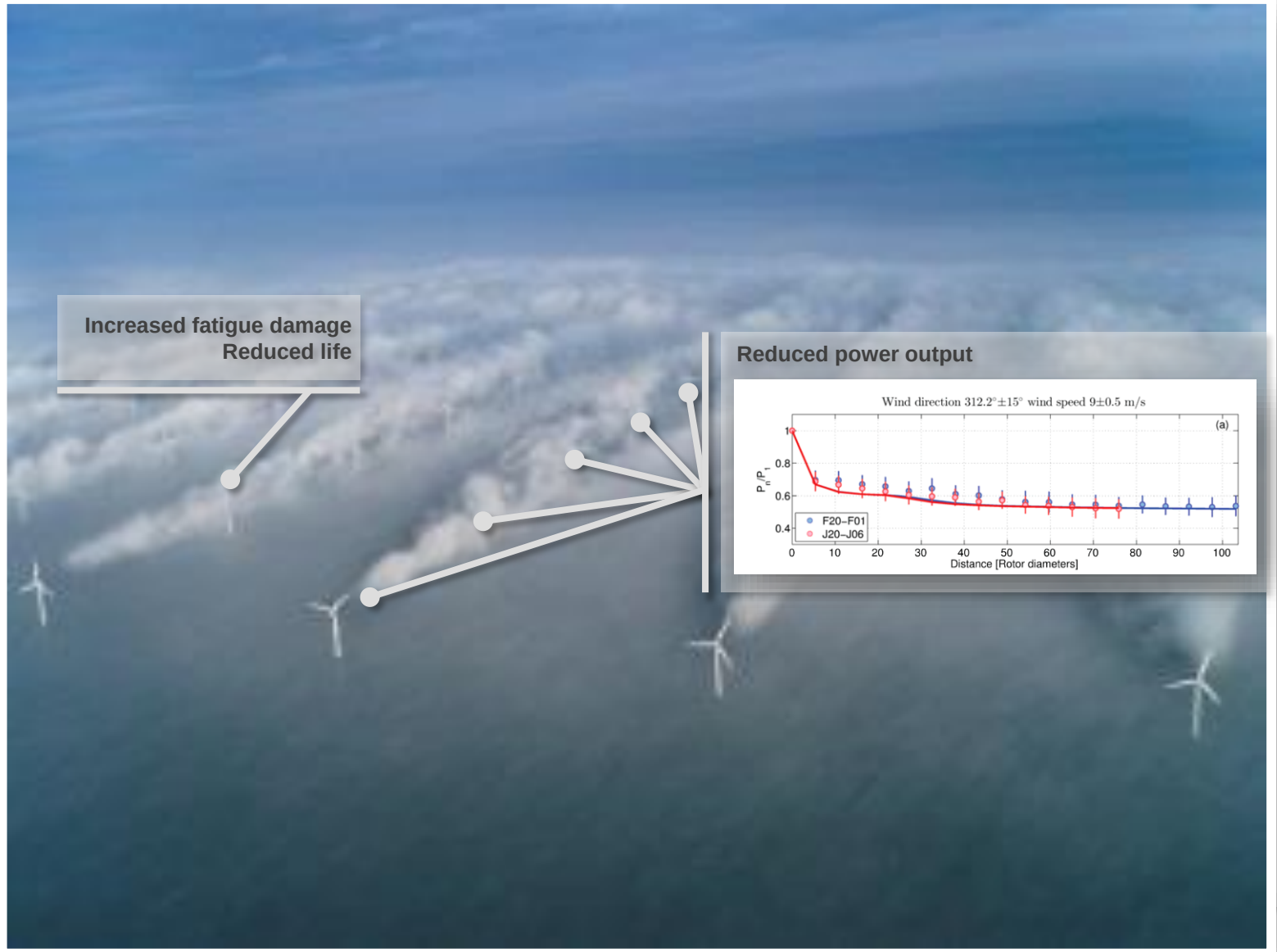
- Areas of Technological Innovation - Wind Plant Level
- Wind Farm Control



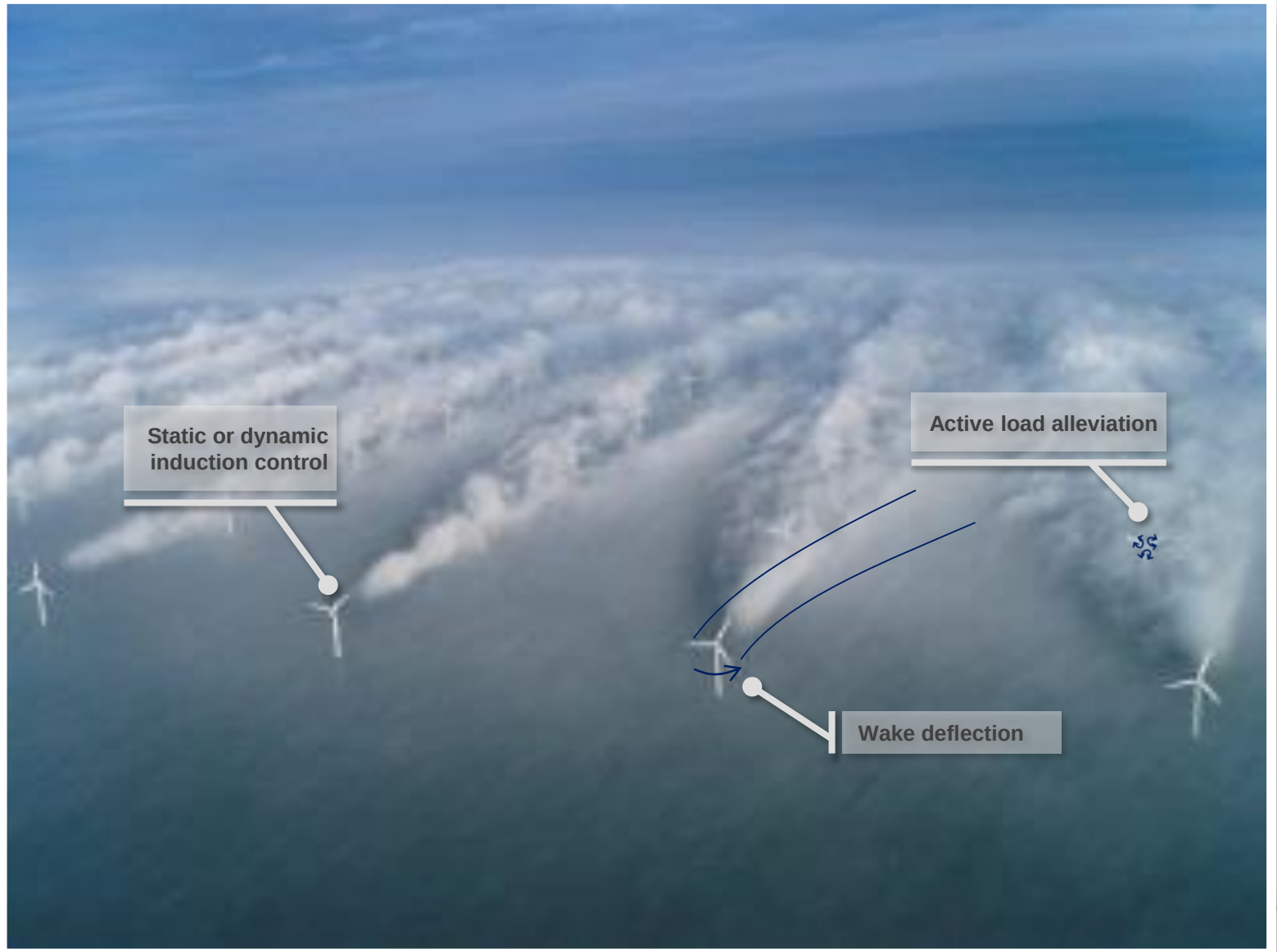
Areas of Technological Innovation - Wind Plant Level -



Wind Farm Effects

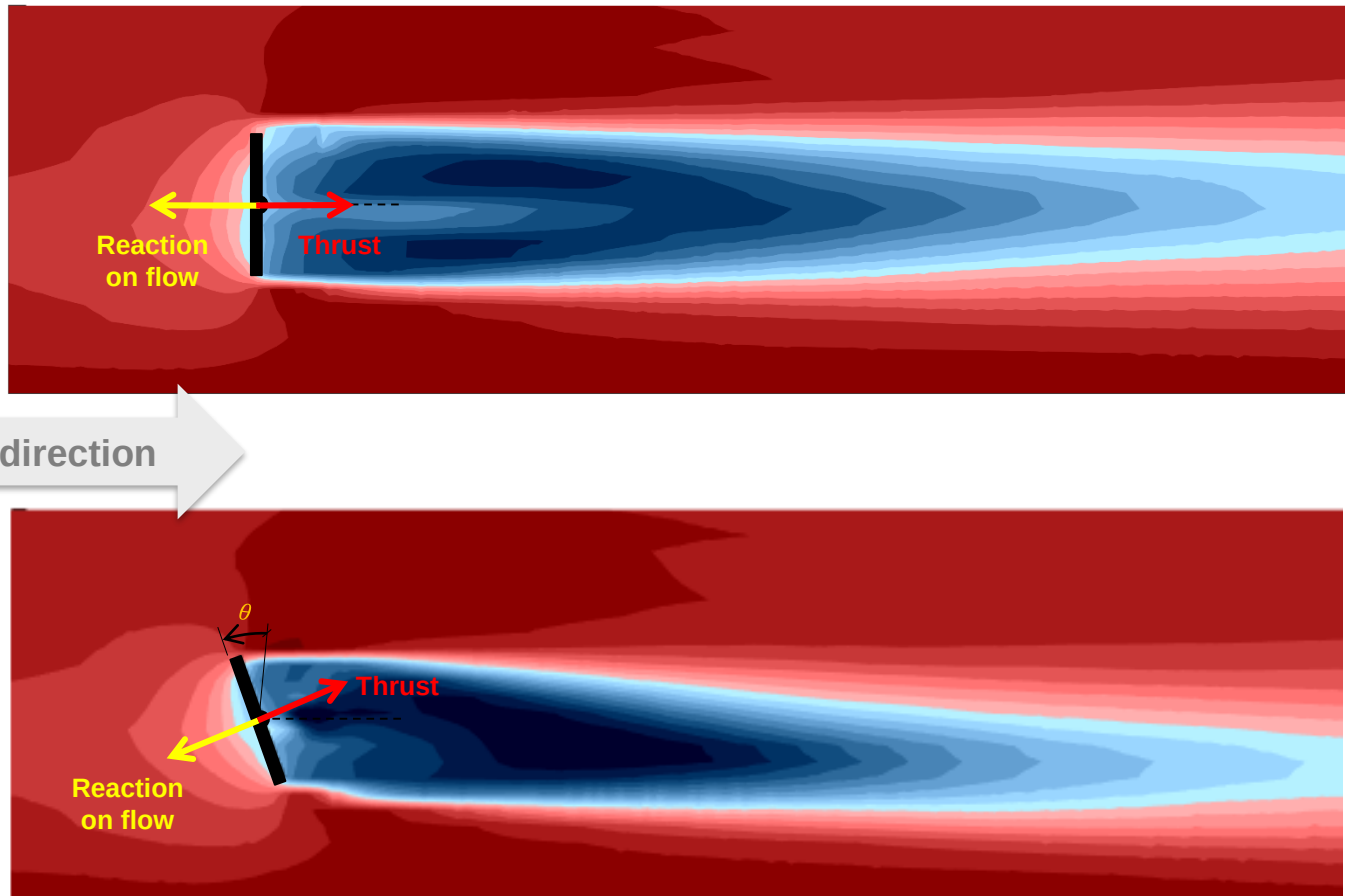


Wind Farm Control



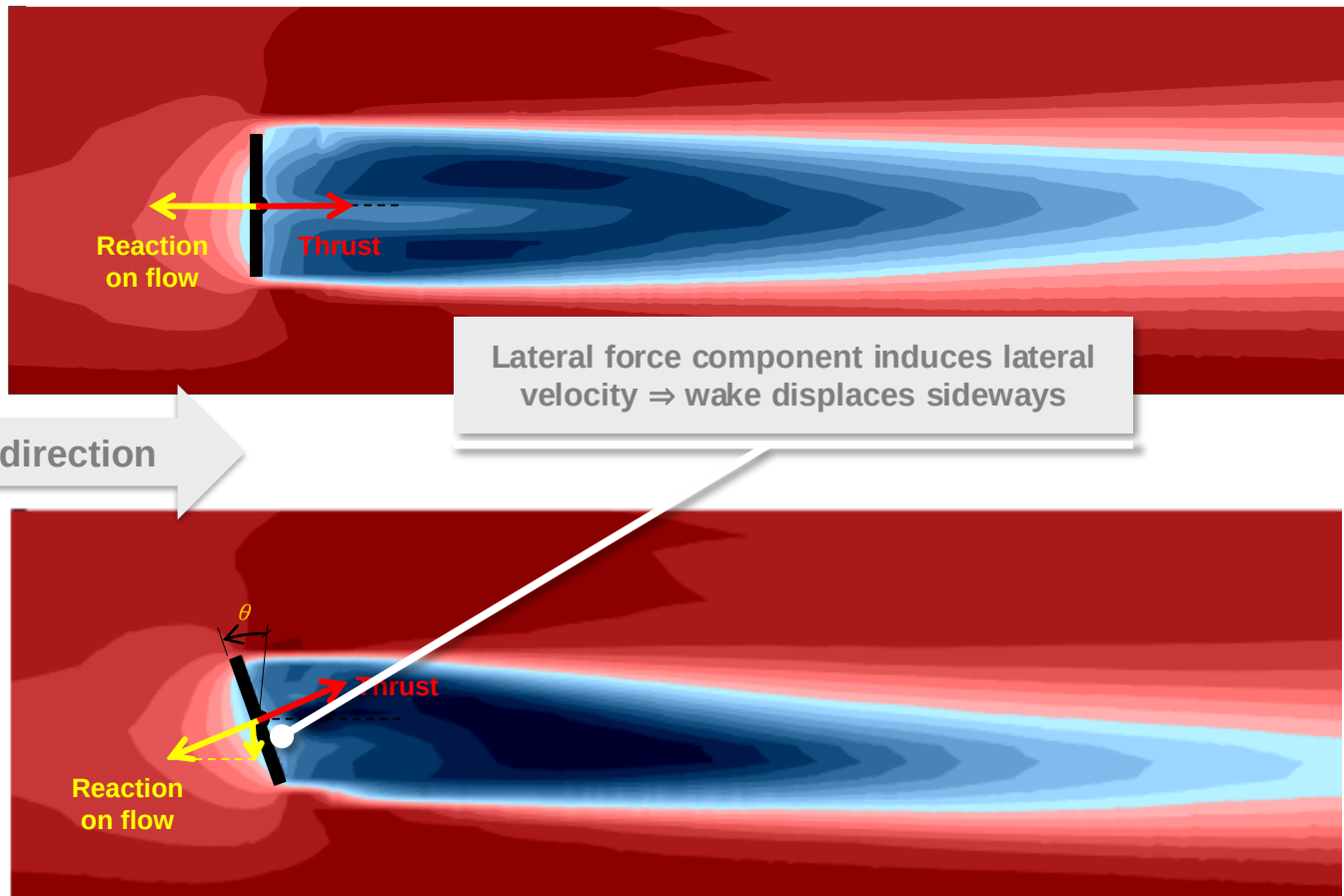
The Concept of Wake Deflection

Yawing a wind turbine out of the wind **deflects its wake:**

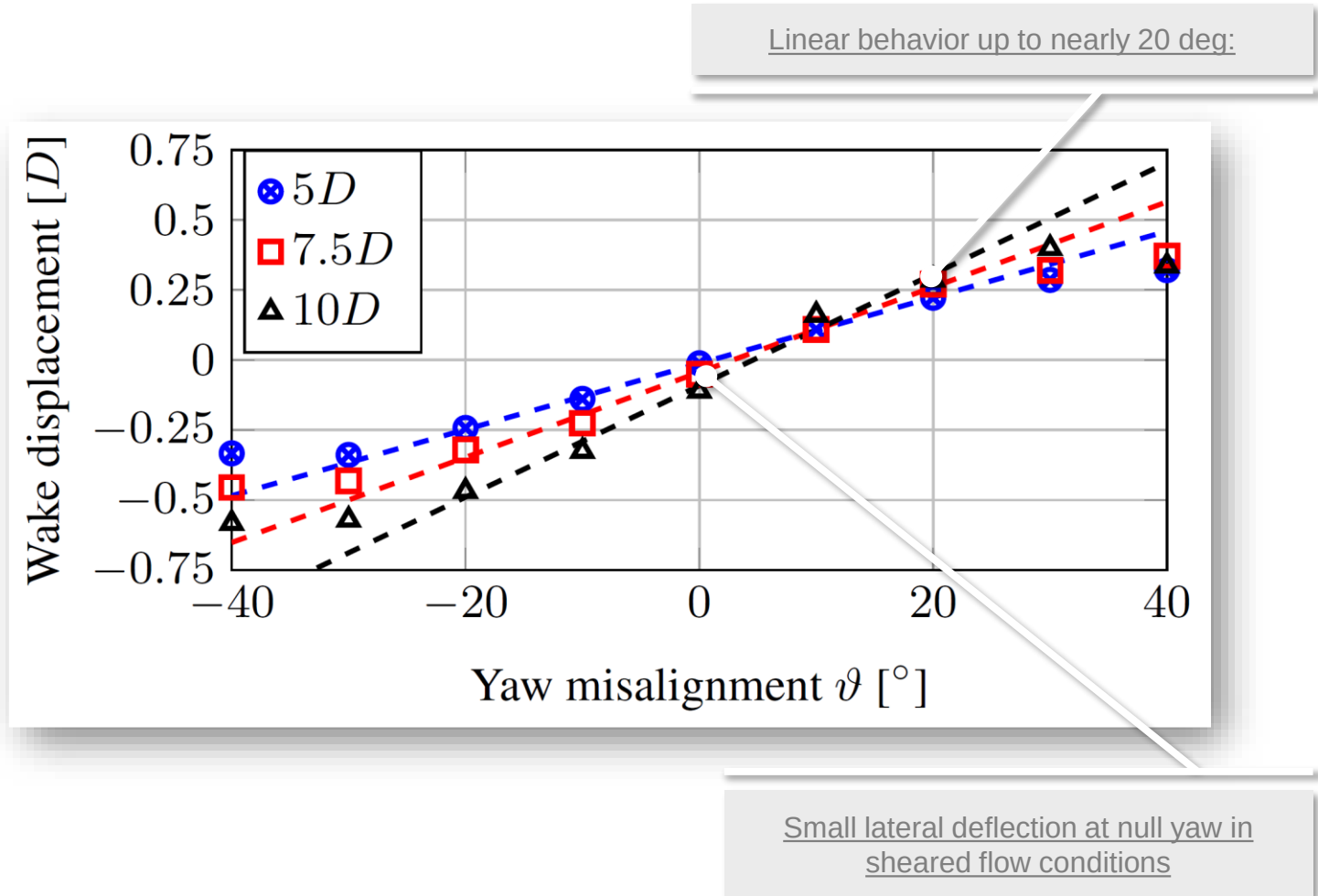


The Concept of Wake Deflection

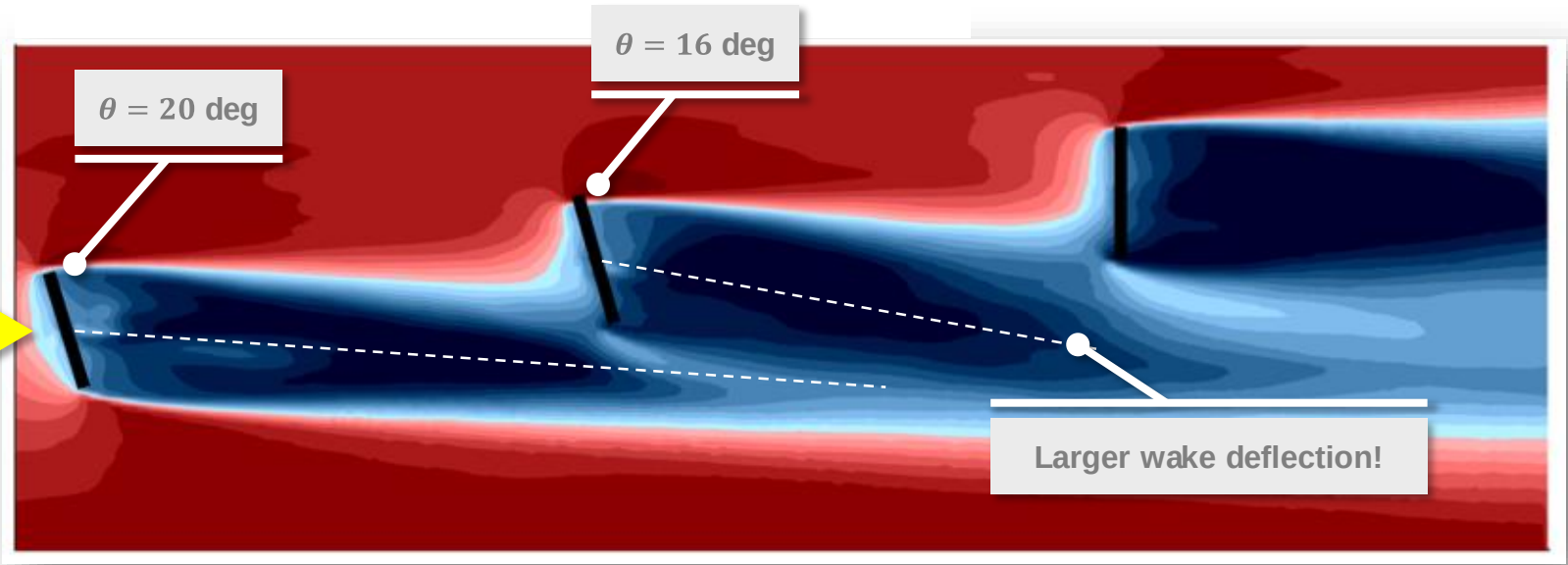
Yawing a wind turbine out of the wind **deflects its wake:**



The Concept of Wake Deflection



Secondary Steering



Why?

Secondary Steering

